

SHARDARA, Kazakhstan

WMO: 384390

Lat: 41.2492N

Lon: 67.9692E

Elev: 275

StdP: 98.06

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-10.7	-16.4	0.9	-13.7	-13.2	1.2	-10.2	9.4	4.0	8.4	3.9	2.1	0	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	38.3	21.3	36.7	20.8	35.3	20.4	24.0	34.2	23.0	33.2	22.0	32.5	3.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	15.8	31.4	19.3	14.5	30.3	18.1	13.5	29.1	73.9	34.1	69.6	33.3	66.0	32.5	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.0	6.0	-15.3	41.5	4.3	1.3	-18.4	42.5	-20.9	43.2	-23.3	44.0	-26.4	45.0		
(5)			-15.7	26.1	4.2	1.5	-18.7	27.2	-21.2	28.0	-23.5	28.8	-26.6	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	0.2	2.2	9.1	15.8	22.2	27.2	29.1	27.6	21.9	14.7	7.4	1.2		
	DBStd	11.17	5.13	5.64	4.70	4.55	3.51	2.95	2.55	2.93	3.83	4.75	5.24	4.96		
	HDD10.0	1001	304	219	74	8	0	0	0	0	0	15	107	273		
	HDD18.3	2412	562	451	288	101	8	0	0	0	12	131	329	531		
	CDD10.0	2807	0	2	45	182	380	515	591	547	356	160	29	1		
	CDD18.3	1175	0	0	0	23	129	265	333	288	118	17	1	0		
	CDH23.3	14189	0	0	2	172	1213	3303	4568	3630	1173	126	3	0		
	CDH26.7	7276	0	0	0	39	438	1708	2637	1977	453	23	0	0		
(14) Wind	WSAvg	2.6	2.5	2.6	2.6	2.6	2.9	2.9	2.8	2.7	2.5	2.3	2.5	2.6		
(15) Precipitation	PrecAvg	408	49	59	61	57	38	13	5	2	5	23	46	51		
	PrecMax	597	76	130	105	133	75	36	26	12	17	76	127	125		
	PrecMin	217	18	5	18	12	0	0	0	0	0	2	6	6		
	PrecStd	86	19	28	22	28	24	9	7	3	5	19	31	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	16.1	22.7	30.1	34.7	39.1	41.0	39.9	35.3	29.3	22.9	13.0		
		MCWB	7.5	10.7	14.1	17.9	19.0	20.8	22.0	22.4	20.1	16.6	14.4	8.4		
	2%	DB	9.6	12.8	20.2	27.6	32.5	36.8	38.8	37.7	32.9	26.5	19.3	11.0		
		MCWB	6.2	8.3	12.6	16.6	18.8	20.7	21.2	21.3	19.1	15.5	12.8	7.4		
	5%	DB	7.7	10.6	18.0	25.2	30.9	35.3	37.2	36.1	31.2	24.3	16.9	9.0		
		MCWB	5.1	7.1	12.0	16.2	18.3	20.1	20.9	20.8	18.5	14.7	11.3	5.9		
	10%	DB	6.2	8.9	15.9	23.0	29.2	33.8	35.6	34.6	29.3	22.4	14.6	7.4		
		MCWB	4.2	6.1	11.1	15.6	18.0	19.7	20.5	20.4	17.8	13.8	10.0	5.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	10.7	15.4	19.6	21.9	24.6	25.6	26.0	22.4	17.9	15.0	9.5		
		MCDB	11.4	15.6	20.4	26.6	30.5	34.3	35.5	35.2	31.9	27.0	21.3	12.1		
	2%	WB	6.5	8.8	13.7	18.3	20.6	23.1	24.0	24.0	20.7	16.5	13.3	7.6		
		MCDB	9.1	12.1	18.4	24.7	28.7	32.8	34.5	33.9	30.3	24.6	18.4	10.1		
	5%	WB	5.4	7.4	12.5	17.1	19.7	22.0	23.0	22.7	19.4	15.4	11.7	6.2		
		MCDB	7.5	10.2	17.1	23.8	28.1	31.8	33.8	32.9	29.5	23.1	16.3	8.6		
	10%	WB	4.3	6.2	11.3	15.9	18.9	21.1	22.0	21.5	18.2	14.3	10.4	5.1		
		MCDB	6.1	8.7	15.4	22.2	27.2	30.9	33.2	32.2	28.4	21.6	14.2	7.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	7.6	9.0	10.1	11.2	12.1	12.7	13.2	13.3	11.5	8.4	6.0		
		MCDBR	7.6	9.4	11.3	13.0	13.3	13.8	14.7	14.5	14.0	13.1	10.2	7.6		
	5% WB	MCWBR	5.3	6.2	6.2	6.1	6.0	5.9	6.3	6.9	6.9	6.7	6.0	5.2		
		MCWBR	7.0	8.6	9.8	11.2	11.4	11.6	12.3	13.1	13.7	12.1	9.2	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.388	0.427	0.480	0.497	0.480	0.469	0.446	0.422	0.412	0.466	0.449	0.389		
	taud		2.070	1.991	1.913	1.921	2.007	2.045	2.117	2.164	2.131	1.928	1.955	2.042		
	Ebn at Noon		717	749	756	779	805	813	829	839	816	694	634	671		
	Edh at Noon		113	143	173	184	173	167	154	142	137	150	127	108		
(44) All-Sky Solar Radiation	RadAvg		1.68	2.57	3.84	5.35	6.80	7.68	7.63	6.86	5.45	3.59	2.15	1.50		
	RadStd		0.20	0.29	0.32	0.52	0.35	0.29	0.24	0.20	0.17	0.25	0.27	0.20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (8 neighbors)	N/A	N/A	N/A	+0.57	N/A	N/A	N/A	N/A	+103	+77

Nomenclature: See separate page